

1 Solution — GIAM 8.2

Problem 3

1.1 Problem

Prove that any two circles are equinumerous (as sets of points).

1.2 Setup — Circles as Sets of Points

A *circle* in $\mathbb{R}^2$ with center $O = (h, k)$ and radius $r > 0$ is the set

$$C = \{ (h + r \cos \theta, k + r \sin \theta) : \theta \in [0, 2\pi) \}.$$

Equivalently, $C = \{ (x, y) \in \mathbb{R}^2 : (x - h)^2 + (y - k)^2 = r^2 \}$.

The map $\theta \mapsto (h + r \cos \theta, k + r \sin \theta)$ is a bijection $[0, 2\pi) \rightarrow C$ (distinct angles in $[0, 2\pi)$ give distinct points, and every point on C is hit by exactly one such angle). Call this map φ_C ; it is the “polar coordinate parametrization” of C .

So *every* circle, regardless of its center or radius, comes with a canonical bijection to the half-open interval $[0, 2\pi)$.

1.3 Strategy — Pair Points by Matching Angle

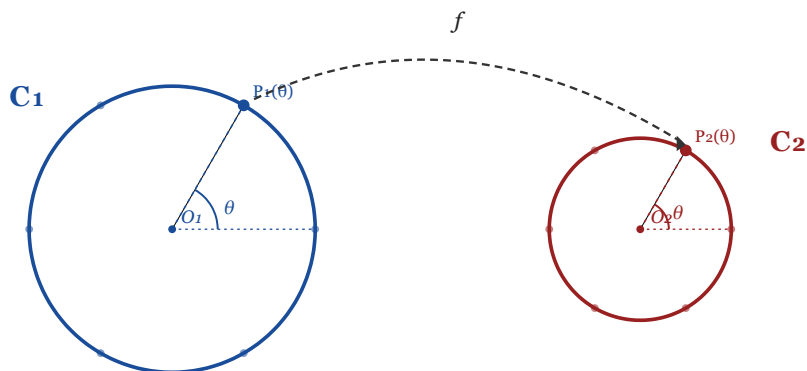
Given two circles C_1 (center O_1 , radius r_1) and C_2 (center O_2 , radius r_2), each has its own angle parametrization $\varphi_{C_1}, \varphi_{C_2} : [0, 2\pi) \rightarrow C_i$. Define $f : C_1 \rightarrow C_2$ as the composite

$$f = \varphi_{C_2} \circ \varphi_{C_1}^{-1}.$$

Explicitly: given $P \in C_1$, find the unique $\theta \in [0, 2\pi)$ with $P = O_1 + r_1(\cos \theta, \sin \theta)$, and set

$$f(P) = O_2 + r_2(\cos \theta, \sin \theta).$$

The map f pairs the point on C_1 “at angle θ from O_1 ” with the point on C_2 “at angle θ from O_2 .”



Same angle θ , different circles: pair $P_1(\theta)$ on C_1 with $P_2(\theta)$ on C_2 .

Figure 1.1: Same angle ϑ , different circles. For each $\vartheta \in [0, 2\pi)$ there is a unique point $P_1(\vartheta)$ on C_1 and a unique point $P_2(\vartheta)$ on C_2 ; the bijection f sends $P_1(\vartheta) \mapsto P_2(\vartheta)$.

1.4 Verifying f is a Bijection

f is a bijection because both φ_{C_1} and φ_{C_2} are bijections from $[0, 2\pi)$ to their respective circles, and the composite of bijections is a bijection (Section 8.1, Problem 2 — transitivity of $\equiv$ via composition).

Explicitly, the inverse is

$$f^{-1} : C_2 \rightarrow C_1, \quad f^{-1}(O_2 + r_2(\cos \theta, \sin \theta)) = O_1 + r_1(\cos \theta, \sin \theta),$$

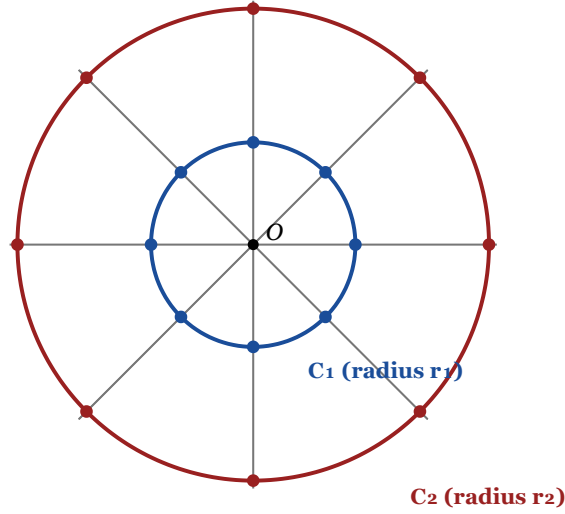
which is $\varphi_{C_1} \circ \varphi_{C_2}^{-1}$ — the same construction in reverse. Hence $C_1 \equiv C_2$. ■

Since C_1 and C_2 were arbitrary circles, this shows *any two circles are equinumerous* — as a corollary, all circles in $\mathbb{R}^2$ belong to a single equivalence class under $\equiv$.

1.5 The Concentric Special Case — Radial Projection

When $O_1 = O_2 = O$, the bijection f has a vivid geometric meaning: each ray emanating from the common center O intersects C_1 at one point and C_2 at one point, and f pairs these two intersection points. Equivalently, f rescales each radius from length r_1 to length r_2 :

$$f(P) = O + \frac{r_2}{r_1}(P - O), \quad P \in C_1.$$



Concentric case: each ray from O picks one point per circle.

Figure 1.2: Concentric circles share a center O . Each ray from O picks one point of C_1 and one of C_2 , so radial projection is the bijection.

This concentric picture is just the angle-parametric bijection drawn with the two centers coinciding — the “angle from O_1 ” and “angle from O_2 ” are then literally the same angle along the same ray.

1.6 Remark — General Case as Translate + Scale

The bijection f between non-concentric circles decomposes into two elementary bijections of $\mathbb{R}^2$, restricted to circles:

$$C_1 \xrightarrow{\text{translate by } O_2 - O_1} C'_1 \xrightarrow{\text{radial scale by } r_2/r_1 \text{ about } O_2} C_2.$$

The first arrow shifts C_1 rigidly so its center moves from O_1 to O_2 ; this produces an intermediate circle C'_1 concentric with C_2 but still of radius r_1 . The second arrow rescales radially about O_2 , stretching (or shrinking) C'_1 to a circle of radius r_2 — namely C_2 . Each arrow is a bijection (translations and central scalings are bijections of $\mathbb{R}^2$); the composite is a bijection by transitivity, recovering f .

This decomposition is the “similarity transformation” view: any two circles in the plane are related by a similarity (translation + scaling), and similarities are bijections of $\mathbb{R}^2$.

1.7 Remark — Cardinality is Blind to Radius and Position

The bijection makes vivid the point that *cardinality* is insensitive to *metric* features. The two circles can differ in:

- *radius* — a one-meter circle has the same number of points as a one-kilometer circle, or as a circle the size of the observable universe;
- *position* — translating a circle anywhere in the plane (or to a different copy of the plane) doesn’t change the cardinality;
- *circumference* — the length of C_1 is $2\pi r_1$ and of C_2 is $2\pi r_2$, but $|C_1| = |C_2|$ regardless.

In fact, every circle has the same cardinality as $[0, 2\pi)$, hence (by the bijection $[0, 2\pi) \rightarrow [0, 1)$ via $\theta \mapsto \theta/(2\pi)$) the same cardinality as any non-degenerate interval, hence — chaining through Problem 2 of this section and the half-circle/line projection — the same cardinality as $\mathbb{R}$. So every circle has cardinality $\mathfrak{c} = |\mathbb{R}|$.

1.8 Remark — The Same Argument Generalizes

The argument hinges on *one* fact: every circle is in bijection with a fixed reference space ($[0, 2\pi)$). Many other geometric objects share this property, and the same composition-via-reference trick applies:

Two objects of the same “type”	Reference parameter space	Bijection witness
--------------------------------	---------------------------	-------------------

two circles	$[0, 2\pi)$	$\varphi_{C_2} \circ \varphi_{C_1}^{-1}$
two line segments	$[0, 1]$	the affine map of Problem 2
two spheres in $\mathbb{R}^3$	$[0, \pi] \times [0, 2\pi)$	spherical-coord composite
two regular polygons (boundaries)	$[0, 1]$	arclength composite
two ellipses	$[0, 2\pi)$	angle-parameter composite

Table 1.1

The lesson is: *whenever two sets admit a parametrization by the same set, they are equinumerous*. The proof above is the cleanest possible instance of this pattern — circles are the simplest shapes with a one-parameter family of points.

1.9 Remark — Connection to “Same Length, Different Length”

Note a small but important contrast with Problem 2 of this section. Problem 2 showed $[a, b] \equiv [c, d]$ for any choice of $a < b$ and $c < d$ — *line segments* of different lengths are equinumerous. The current problem says *circles* of different radii are equinumerous. Both reduce to the same idea — “rescale the parameter” — but the geometric flavor differs:

- For segments, the bijection $x \mapsto c + (x - a)(d - c)/(b - a)$ is an affine *stretch* along a single dimension.
- For circles, the bijection wraps a fixed parameter θ around two circles of different sizes — a “rotational rescaling” rather than a linear one.

Both, however, are instances of: pick a common parameter space and pull bijections through it.